

USB CLI Reference

Central Control Hub (Pico 2 W master)

RS-485 master + WiFi gateway on Pico 2 W. The `wifi` command is overridden by the adapter (CYW43 stack) and shadows the core ESP8285 implementation. Optionally builds with BUILD_TYPE_WATER_DISPENSER for a co-located dispenser. Adapter file: application/Source/Adapters/USB_Command_Adapter.c

Source: sevantica_drivers/Source/USB/USB_Command_Handler.c (core) + project adapter application/Source/Adapters/USB_Command_Adapter.c. Adapter commands are dispatched *before* core commands, so an adapter may shadow a core command of the same name.

help / ?

Origin: core

Show available top-level commands (adapter + core).

Command	Description	Example
help	List all registered commands with descriptions.	help
?	Alias of `help`.	?
clear	Clear the terminal screen and re-print the welcome banner.	clear

sys

Origin: core

System-level inspection and lifecycle control.

Command	Description	Example
sys status	Print device ID, site ID, RTC, MIFARE state, free heap, etc.	sys status
sys config	Dump the full live configuration (`Config_PrintToUSB`).	sys config
sys set <param> <value>	Set a single configuration parameter and persist to SD + flash.	sys set mifare.card_init_default_balance 5000
sys version	Show firmware version and build metadata.	sys version
sys reboot	Software reset via watchdog.	sys reboot
sys bootloader	Reboot into RP2040 BOOTSEL (USB mass-storage) mode for re-flashing.	sys bootloader
sys tasks	Print FreeRTOS task list with min-free stack (words).	sys tasks

Command	Description	Example
sys wdt [save erase raw]	Watchdog log: dump latest entries; `save` flushes RAM->flash, `erase` wipes the WDT sector, `raw` dumps the raw flash slots.	<pre>sys wdt sys wdt save sys wdt erase</pre>
sys security <on off>	Enable / disable the full MIFARE security stack (encryption, HMAC, replay, challenge-response, custom sector keys).	<pre>sys security on</pre>
sys factory-reset	Reload immutable defaults from the flash factory sector.	<pre>sys factory-reset</pre>
sys factory-save	Snapshot current source defaults into the flash factory sector.	<pre>sys factory-save</pre>
sys defaults-apply	Overwrite live (user) settings with the immutable source defaults.	<pre>sys defaults-apply</pre>

mod

Origin: core

Runtime control of system modules (start / stop / list).

Command	Description	Example
mod list	Print module table with current run state.	<pre>mod list</pre>
mod start <module>	Start a module. Known names: lcd, mifare, dispenser, buzzer, ioexp, rs485.	<pre>mod start dispenser</pre>
mod stop <module>	Stop a module by name (same names as `mod start`).	<pre>mod stop buzzer</pre>

card

Origin: core | Build flag: USE_DRIVERS_NFC

MIFARE card maintenance and recovery (gated by USE_DRIVERS_NFC).

Command	Description	Example
card format	Initialize / re-format the next presented MIFARE card with default data.	<pre>card format</pre>
card topup <amount>	Add tokens / volume units to the next presented card.	<pre>card topup 5000</pre>
card log <UID_HEX>	Read and print the on-card transaction log for the given UID.	<pre>card log 04A1B2C3D4</pre>
card recover <UID_HEX>	Restore card balance from the SD-side recovery log for that UID.	<pre>card recover 04A1B2C3D4</pre>
card decrypt	Decrypt and dump the next presented card (10 s wait) using current keys.	<pre>card decrypt</pre>

fs

Origin: core | Build flag: USE_DRIVERS_FATFS

SD card file system access (gated by USE_DRIVERS_FATFS).

Command	Description	Example
fs ls	List files in the SD root and cache them for `fs cat #N` access.	<code>fs ls</code>
fs cat <name #N>	Print contents of a file by name, or by index from the last `fs ls`.	<code>fs cat config.txt</code> <code>fs cat #2</code>
fs find <UID_HEX>	Search SD logs for entries belonging to that card UID.	<code>fs find 04A1B2C3D4</code>
fs format	Delete every file on the SD card (does NOT low-level format).	<code>fs format</code>
fs mount	Expose the SD card to the PC via USB Mass Storage; suspends application I/O.	<code>fs mount</code>
fs unmount	Detach USB MSC and resume application access to the SD card.	<code>fs unmount</code>

rtc

Origin: core

Real-time clock inspection and manual set.

Command	Description	Example
rtc status	Print current time, sync flag, unix timestamp and compile-time fallback.	<code>rtc status</code>
rtc set <YYYY-MM-DD> <HH:MM:SS>	Set RTC date/time and persist to SD.	<code>rtc set 2026-05-04 14:30:00</code>

dispenser

Origin: adapter | Build flag: BUILD_TYPE_WATER_DISPENSER

Local dispenser control (only in BUILD_TYPE_WATER_DISPENSER builds of CCH).

Command	Description	Example
dispenser start [L]	Manual dispense, identical semantics to MyWota.	<code>dispenser start 5</code>
dispenser stop	Stop manual dispense.	<code>dispenser stop</code>
dispenser topup <L>	Top up the card by N litres.	<code>dispenser topup 10</code>

wifi (CCH adapter)

Origin: adapter

CYW43-based WiFi via the Network task. Overrides the core ESP8285 `wifi` because adapter commands are dispatched first.

Command	Description	Example
wifi scan	Start a background scan (results retrieved by `wifi connect <ID>`).	wifi scan
wifi connect <ID> <pwd>	Connect using the scan result index from the last scan.	wifi connect 1 myPassword
wifi connect <pwd>	Connect to the compiled-in DEFAULT_WIFI_SSID using <pwd>.	wifi connect myPassword
wifi disconnect	Disconnect from the current network.	wifi disconnect
wifi saved	List saved networks (auto-connect MRU list, max WIFI_MAX_SAVED_NETWORKS).	wifi saved
wifi forget <index>	Remove a saved network by index (from `wifi saved`).	wifi forget 0
wifi reinit	Re-initialize the WiFi driver.	wifi reinit
wifi status	Print network state and IP (refreshes IP if 0.0.0.0).	wifi status

sdinit

Origin: adapter

Re-run SD card initialization (CCH adapter).

Command	Description	Example
sdinit	Request the SD logger task to re-mount and re-initialize the card.	sdinit

rs485 / remote

Origin: adapter

RS-485 master bus inspection and slave CLI bridging (CCH adapter).

Command	Description	Example
rs485	Print the RS-485 slave table (address, online, type, state, balance, volume).	rs485
remote <addr> <command...>	Send a CLI command string over RS-485 to a slave; the slave runs it locally.	remote 2 sys status remote 5 dispenser stop

webserver

Origin: adapter

HTTP server lifecycle (CCH adapter).

Command	Description	Example
webserver start	Start the embedded web server (requires WiFi connected).	webserver start

Command	Description	Example
webserver status	Print WiFi connection state and the server IP.	<code>webserver status</code>

clean (master)

Origin: adapter

Site-wide clean orchestration over RS-485 (CCH adapter, different semantics from MyWota's local `clean`).

Command	Description	Example
clean now <addr> [vol_ml] [max_sec]	Trigger an immediate clean on a specific slave dispenser.	<code>clean now 2 2000 60</code>
clean abort <addr>	Abort an in-progress clean on a slave.	<code>clean abort 2</code>
clean status	Print scheduler state for all slaves.	<code>clean status</code>

filter

Origin: adapter

Site-shared filter life accounting (CCH adapter).

Command	Description	Example
filter status	Print capacity, used, remaining %, session totals.	<code>filter status</code>
filter reset	Zero the used counter (after a physical filter replacement).	<code>filter reset</code>
filter set <capacity_ml>	Set the filter capacity in mL (0 disables tracking).	<code>filter set 50000</code>

stats

Origin: adapter

Per-slave throughput statistics, hourly buckets (CCH adapter).

Command	Description	Example
stats	Dump the rolling per-slave throughput buckets.	<code>stats</code>

Missing / Inconsistent Commands

Cross-checked against: MyWota (Pico water dispenser), BigYellow (Pico car-wash terminal).

`fault` — missing in Central Control Hub (Pico 2 W master)

Present in: MyWota (Pico water dispenser).

`washstart` — missing in Central Control Hub (Pico 2 W master)

Present in: BigYellow (Pico car-wash terminal).

`washstop` — missing in Central Control Hub (Pico 2 W master)

Present in: BigYellow (Pico car-wash terminal).

`fault` — missing

MyWota has `fault status|clear` for its dispenser. CCH could surface aggregated slave fault state (per-slave or master).

`clean` semantics differ from MyWota

On CCH, `clean now/abort/status` orchestrates slaves over RS-485. On MyWota, `clean start/stop/status` is local. The verb overlap is intentional but easy to confuse — document carefully.